

object, and the high-resolution object is the base object. A base object can be any type of deformable object.

6.6 Morphing

Morphs are managed in 3ds Max by using the Morpher modifier. This allows you to change the shape of one object to match that of another. This is a precise way to deform an object, as each target of the Morpher modifier can be modeled by using 3ds Max's modeling tools. Morphs can also combine multiple targets to mix and match deformations. The big task in configuring morphs is creating the targets. This is primarily a modeling task. The one requirement for blend shape targets is that they all have the same topology as the base object. This is easily accomplished by duplicating the base object (such as a head) and reshaping it to create the appropriate shape (such as a blink or a smile.)

After the targets are created, setting up the morphs is simply a matter of adding the Morpher modifier to the stack and then selecting the targets by using the tools in the modifier. Each target gets a channel, which allows you to set the morph amount for each target. Morphs work by averaging the X, Y, and Z positions of each individual vertex or CV in the object. When a slider is at 0 percent, the vertex is at the rest position; at 100 percent, it is at the target. Mixing multiple targets averages the positions according to the weights on the sliders.

6.6.1 Facial animation using Morphs

Although morphs can be used for any type of shape animation, by far the most popular application for blend shapes is in facial animation. A modeler creates individual facial poses representing the extreme motions of the individual parts of the face, such as opening the jaw or a smile. By mixing these, you can create an infinite variety of facial expressions.

"A" and "I" / "Eh" and "Uh" / "F" and "V" / "L" and "Th" / "M" "Oooh" / "Oh" "E" and consonants, Blend shapes called phonemes can be created



Facial expressions

to mimic the basic mouth positions used in dialogue. Other blend shapes can be modeled for positions the mouth makes when not speaking, such as a smile. It's always a good idea to create right and left versions